Faisal Ahmed

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Google Scholar | *GitHub*

Research Interest

- IoT systems, edge AI, and security.

Education

Bachelor of Science in Computer Science and Engineering **May/2026**
East West University, Dhaka 1212

- Result CGPA 3.73/4.00
- Capstone research: PAUMIoT, supervised by Hasan Mahmood Aminul Islam; published at DCOSS-IoT 2026.

Higher Secondary School Certificate (HSC) **January/2021**
Dhaka College, New Market, Dhaka 1205

- GPA 5.00/5.00 in all subjects. Average marks: 96%

Research Experience

Capstone Research, IoT Systems **2025 – Present**
East West University, Dhaka. Supervised by Hasan Mahmood Aminul Islam

- Built the ESP32 client for PAUMIoT, which picks MQTT, CoAP or HTTP at runtime based on device and network conditions.
- Added a safety shield so the device falls back to a safe protocol when its battery or signal gets too low.
- Set up the ESP32 and sensor testbed and ran all the experiments.
- Wrote the communication layer and core algorithms of SEFIoT.
- Extended The ONE simulator with a CCN layer, and added a publication register to measure how often cached content was already out of date.

Independent Research, IoT Attack Measurement **Apr 2026 – Present**
Honeypot study; journal paper in preparation

- Ran Cowrie, Dionaea, Glutton and OpenCanary on one cloud server, with each sensor listening on its own ports.
- Collected about 4.0 million attack events from 29,052 IPs over 112 days.
- Compared attacker IPs with Shodan and Censys records of 51,738 IoT devices and with threat-feed data.

Publications

Conference papers

1. H. M. A. Islam, M. M. R. Maharaz, **Faisal Ahmed**, M. F. H. Dipto, M. M. H. Shafin, and M. Georgiades, "Towards a protocol-agnostic universal middleware for IoT with edge intelligence," in *Proc. 22nd Int. Conf. Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT)*, 2026, doi: *10.1109/DCOSS-IoT69657.2026.00046*.
2. M. M. Rahman, **Faisal Ahmed**, M. F. H. Dipto, M. M. H. Shafin, S. M. N. Shahriar, P. Akibuzzaman, and H. M. A. Islam, "Emulating sensor profile towards IoT simulator development," in *Proc. IEEE 2nd Int. Conf. Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, doi: *10.1109/QPAIN69676.2026.11546398*.
3. F. R. Sams, S. K. Supti, M. J. Ali, **Faisal Ahmed**, M. F. H. Dipto, R. Gani, M. B. N. Shuci, M. R. A. Rashid, and S. H. Ripon, "Interpretable machine learning for early diabetes prediction with statistical validation and explainable AI," in *Proc. IEEE Int. Conf. Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2025, doi: *10.1109/BECITHCON69222.2025.11504208*.

Submitted:

1. M. M. Rahman, **Faisal Ahmed**, M. F. H. Dipto, and M. M. H. Shafin, "Freshness before the cache: Measuring content staleness at ingress in CCN over delay-tolerant networks," submitted to *13th Int. Conf. Next Generation Computing, Communication, Systems and Security (NSysS)*, 2026.

Journal papers

In Preparation:

1. **Faisal Ahmed**, M. M. R. Maharaz, M. F. H. Dipto, M. M. H. Shafin, and M. Georgiades, "PAUMIoT: Safety-shielded, federated, cost-aware transport adaptation for heterogeneous IoT."
2. **Faisal Ahmed**, N. Yeasmin, R. Hossen, and M. Georgiades, "Passive attribution of role-separated IoT attack ecosystems through orthogonal behavioral fingerprints."

Skills

- **Hardware and embedded:** ESP32 (FreeRTOS), sensors, hardware-in-the-loop testing
- **Security and measurement:** Cowrie, Dionaea, Glutton, OpenCanary, Shodan, Censys, threat-intelligence feeds
- **Languages:** C, Go, Python, C#
- **Systems and data:** Linux servers, PostgreSQL, The ONE simulator

Teaching Experience

Instructor **01/10/23 - 01/03/24**
Step to Backend, Online sessions

- Taught backend basics in online sessions, mainly databases and how to deploy an app.

Honors and Awards

- Medha Lalon(Dean's) Fund Scholarship
- Talent pool Scholarship in PEC and JSC
- Dhaka Reporters Unity Result Excellency Award given by former **Education Minister Nurul Islam Nahid**

Experience

Co-Founder & Software Engineer **25/07/2025 - Present**
Trelet, Dhaka, Bangladesh

- Wrote the product variant system in Go, PostgreSQL and Next.js, which handles products with many options, tiered prices and stock checks. Also fixed a query that was duplicating rows.
- Set up Stripe payments with signed webhook checks, and wrote 25+ tests that try to tamper with webhooks and payment intents.
- Built role-based access with three roles and 23 security tests. While auditing the code, found a hard-coded secret that allowed a login bypass and removed it.

Full-stack Developer (Remote) **01/10/2025 - 31/12/2025**
RN Jobforce, Shropshire, England

- Worked on the Django backend, including a WebSocket live chat (under 100 ms latency) and Celery notifications.
- Built an interview module on the OpenAI API, which generates questions and gives candidates feedback during the session.

Software Engineering Intern **01/04/2025 - 30/09/2025**
Progress IT Solution, Dhaka, Bangladesh

- Wrote a C# video processor using FFmpeg and OpenCV, with 15 custom shaders and frame interpolation.
- Built a Windows desktop app (WPF/WinForms) for a cybersecurity client, which automates domain registration and Group Policy setup.